

FitHub

Smart Gym & Wellness Management System



Software Test Specification (STS)
Based on IEEE 829 – Adapted for Student Projects
Supervisor: Ms. Maha Asiri

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1. Introduction

This document presents **the Software Test Specification (STS) for the FitHub** system.

It describes the document objectives, testing scope, and overall background of the project. In addition, it lists the main definitions, acronyms, and references used throughout this document for the **FitHub** system, which is a **web-based Smart Gym & Wellness Management System** that manages members, trainers, sessions, and workout tracking.

1.1 Objectives

The Software Test Specification (STS) for **FitHub** defines what will be tested, how it will be tested, and who will perform the tests. It ensures that the system is checked against the SRS requirements so that FitHub works as expected.

1.2 Scope

This document defines the scope of testing for the FitHub system. All implemented functions will be tested to ensure that each component behaves as intended. The testing will be performed at the following levels:

- Unit testing: Verify individual backend functions, API endpoints, and key frontend components.
- Integration testing: Check the interaction between the web UI, the Flask backend, and the SQLite database.
- System testing: Execute end-to-end scenarios across different roles (e.g., Member logs in → books a session → attendance is recorded → progress is viewed).
- Acceptance-style testing: Confirm that the main requirements from the SRS and SDD are satisfied for Members, Trainers, and Admins.

These testing levels together ensure that the system is implemented correctly and delivers the expected functionality to its users.

1.3 Reference Material

1. IEEE Standard for Software Test Documentation, IEEE Std 829-1998.
2. Somerville, Ian, "Software Engineering", Addison Wesley (10th edition): 2016, ISBN:10:1- 292-09613-6, ISBN-13:978-1-292-09613-1.

2. Test Items

2.1 Authentication & Navigation

1. Login
2. Logout
3. Registration
4. Forgot Password (System Administrator Request)
5. Role-based access control and navigation (Member / Trainer / Admin)

2.2 Admin Module

1. View user lists (Members & Trainers)
2. Add New Users (Member/Trainer)
3. Activate/Deactivate (Toggle Status) user accounts
4. Manage basic account information (Edit user details)
5. Assigning Trainers to Members
6. View system statistics and admin dashboard data

2.3 Member Module

1. Access the member dashboard
2. Log workouts with required details (e.g., type, duration, calories)
3. View workout history and basic statistics
4. View progress and analytics
5. Browse available sessions
6. Book / Cancel session
7. View and Edit profile information

2.4 Trainer Module

1. Trainer Login and Session Persistence.

2. View Trainer Dashboard.
3. View Trainer Profile.
4. Create and view training sessions.
5. View member list and aggregated statistics (KPIs: Total Members, Avg. Attendance, New This Month).
6. Add New Members.
7. Search and View detailed member information.
8. View performance reports.

2.5 Shared Components

1. UI navigation and consistency
2. Form validation (e.g., required fields, invalid input formats)
3. API error handling (appropriate error responses for invalid input, unauthorized access, and not-found resources).
4. Database interaction through Flask + SQLite

3. Features to Be Tested

3.1 Authentication and Authorization

- User login and logout.
- Role-based access control for Member, Trainer, and Admin.
- Restricting access to pages and actions based on the current role.

3.2 Member Features

- Viewing and updating the member profile.
- Logging workouts with required details (e.g., type, duration, calories).
- Viewing workout history and basic statistics (e.g., total workouts, calories, weekly/monthly/overall summaries).
- Booking and cancelling training sessions.

3.3 Trainer Features

- Creating and managing training sessions.
- Managing and monitoring gym members.
- Analyzing member progress and attendance through performance reports.

3.4 Admin Features

- Viewing the list of users by role (members, trainers).
- Activating and deactivating user accounts.
- Managing basic account information related to users.

3.5 Session and Attendance Management

- Booking a session correctly and cancelling it.
- Recording attendance for sessions.
- Viewing attendance information for sessions or members.

3.6 General UI and Navigation

- Navigating between main pages and dashboards without errors or broken links.
- Displaying clear success and error messages after user actions.
- Basic validation on forms (required fields, invalid input formats).

4.Features Not to Be Tested

Table 1: Features Not to Be Tested

Feature	Reason
Integration with wearable devices (e.g., smartwatches)	Not included in this release of the software.
Online payment or subscription billing	Payment and billing are out of scope for the current project.
Native mobile applications (Android / iOS version of FitHub)	The current release focuses only on the web application.
Advanced analytics or AI-based workout recommendations	Not included in this release of the software.
Automatic email based “Forgot Password” reset (send reset link/ code to user email)	This feature is not implemented in the current version. The system only sends requests to the administrator.
Edit and delete training sessions	These features were planned but not implemented in the current version due to time limitations (only create/view are available).

5. Test Approach

This section describes the overall approach used to test the FitHub system.

5.1 Testing Strategy

The FitHub system was tested using a **manual black-box testing** approach. This strategy focuses on verifying system functionality by providing inputs and observing outputs without examining the internal source code or implementation details. The primary goal

of this approach is to ensure that the system behaves as expected according to the defined functional requirements.

Manual testing was selected due to the scope of the project and the need to validate user interface behavior and usability. Both valid and invalid inputs were used during testing to verify correct system responses, error handling, and input validation for **Member, Trainer, and Admin** workflows.

5.2 Testing Levels

Testing was conducted at multiple levels to ensure adequate coverage of the FitHub system:

- **Unit Testing:** Individual features were tested independently to verify correct functionality. This includes testing member-related functionalities such as login, workout logging, viewing workout history, and session booking session booking, as well as basic trainer and admin actions.
- **Integration Testing:** Integration testing was performed to verify correct interaction between system components, particularly the communication between the frontend and backend. This ensured that data entered through the user interface was correctly processed and returned by the backend services.
- **System Testing:** System testing was conducted to validate the complete FitHub system as a whole. End-to-end user scenarios were executed, such as logging in as a member, navigating through pages, logging workouts, and booking or canceling sessions, to ensure the system operates correctly in a real usage environment.

5.3 Testing Tools

The following tools were used during the testing process:

- Visual Studio Code as the development and execution environment
- System terminal / command line for running backend and frontend services
- Web browsers such as Google Chrome for accessing and testing the system through a localhost environment

Testing results were evaluated through visual inspection by comparing the actual system behavior with the expected outcomes defined in the test cases.

6. Test Cases

6.1 Authentication & Navigation Test Cases

Table 2: Authentication & Navigation Test Case

Test Case ID	Feature	Input / Steps	Expected Output	Actual Result	Pass/Fail
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TC-AUTH-01	Login –valid credentials	1. Open the login page for a specific role (Admin / Trainer / Member). 2. Enter a valid email and the correct password for that role. 3. Click on “Login”.	• User is logged in successfully. • User is redirected to the correct dashboard for their role (Admin / Trainer / Member).	Same as expected results.	Pass
TC-AUTH-02	Login – invalid password	1. Open any login page (Admin / Trainer / Member). 2. Enter a valid email and a wrong password. 3. Click on “Login”.	• Login is rejected. • An error message such as “Invalid username or password” is displayed. • User stays on the login page and is not logged in.	Same as expected results.	Pass
TC-AUTH-03	Forgot Password	On the login page, click on “Forgot Password”.	• A request is sent to the system administrator by the user • A confirmation message appears to the user (Please contact the administrator via email to reset your password).	Same as expected results.	Pass
TC-AUTH-04	Registration – valid data	1. From the login page, click on “Create Account” to open the registration page. 2. Fill in all required fields with valid data (full name, email, password). 3. Click on “Creat Account”.	• A new member account is created and stored in the system. • The user is automatically logged in and redirected to the appropriate interface according to their role (Member dashboard or Trainer dashboard).	Same as expected results.	Pass
TC-NAV-01	Navigation between main pages	1. Login as a valid user (any role: Member / Trainer / Admin). 2. From the dashboard, use the navigation menu to open different main pages (e.g., Dashboard, Sessions, Workouts, Profile).	Each menu item opens the correct page with no broken links or navigation errors.	Same as expected results.	Pass

TC-NAV-02	Logout	1. Login as any valid role. 2. Click on the “Logout” button .	User is logged out, session is cleared, and user is redirected to the login page. The user cannot access protected pages using back button.	Same as expected results.	Pass
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6.2 Admin Test Cases

Table 3: Admin Test Cases

Test Case ID	Feature	Input / Steps	Expected Output	Actual Result	Pass/Fail
TC-ADM-01	Admin – view trainers and members	1. Login as Admin. 2. From the sidebar, click “Manage Trainers”. 3. Observe the trainers table. 4. From the sidebar, click “Manage Members”. 5. Observe the members table.	• On Manage Trainers, a table of trainers is displayed with correct information (Trainer Name, Email, Specialty, Status, Actions). • On Manage Members, a table of members is displayed with correct information (Member Name, Plan, Status, Assigned Trainer).	Same as expected results.	Pass
TC-ADM-02	Admin – add new user (trainer/member)	1. Login as Admin. 2. Open “Manage Trainers” (for adding a trainer) or “Manage Members” (for adding a member). 3. Click on “Add Trainer” or “Add Member”. 4. Fill in all required fields with valid data. 5. Click “Save” / “Add”.	• A new trainer/member record is created and stored in the system. • A success message is displayed (“User added successfully.”). • The new user appears in the corresponding table with correct details. • The Status of the newly added user is set to Active by default.	Same as expected results	Pass

TC-ADM-03	Admin – edit user information	1. Login as Admin. 2. Open “Manage Trainers” or “Manage Members”. 3. In the table, choose an existing user and click “Edit”. 4. Modify one or more fields (for example: plan, specialty, assigned trainer...). 5. Click “Save”.	• The user’s information is updated in the system. • A success message is displayed (“User updated successfully.”). • The table shows the new values correctly.	Same as expected results	Pass
TC-ADM-04	Admin – change user status (activate / deactivate)	1. Login as Admin. 2. Open the “Manage Trainers” page from the sidebar. 3. In the trainers table, choose a user with Status = active and click “Deactivate”. 4. Observe the Status field and the actions in the table. 5. Open the “Manage Members” page. 6. In the members table, choose a user with Status = inactive and click “suspended”. 7. Observe the Status field and the actions in the table.	• For the trainer: -The trainer’s Status is updated from active to inactive. -The status badge changes to the inactive style. -The trainer remains visible in the table with the updated status. • For the member: -When suspending: Status is updated from active to suspended, and the badge shows suspended (e.g., red). -When activating: Status is updated from suspended to active, and the badge shows active (e.g., green). -The member remains visible in the table with the updated status.	Same as expected results	Pass

6.3 Trainer Test Cases

Table 4: Trainer Test Cases

Test Case ID	Feature	Input	Expected Output	Actual Result	Pass/Fail
TC-T01	Landing page Load	User opens the FitHub web app	Landing page loads showing logo, app title, and "Get Started / Trainer Login" buttons	Same as expected result	Pass

TC-T02	Login (valid credentials)	1. Open the login page. 2. Enter Email: testtrainer@test.com Password: password123 3. Click "Login".	System authenticates trainers and redirects to Trainer Dashboard, and the Dashboard loads successfully	Same as expected result	Pass
TC-T03	Login (invalid password)	1. Open the login page. 2. Enter Email: testtrainer@test.com Password: wrong123. 3. Click "Login".	Error 401 Unauthorized	Same as expected result	Pass
TC-T04	Login (invalid email)	1. Open the login page. 2. Enter Email: wrong@example.com Password: password123. 3. Click "Login".	Validation error 400 (Invalid email format)	Same as expected result	Pass
TC-T05	Login (empty field)	1. Open the login page. 2. Leave Email and Password empty. 3. Click "Login".	Client-side validation: required fields	Same as expected result	Pass
TC-T06	Trainer Dashboard	Click on "Dashboard" tab	System displays KPIs, upcoming sessions, and quick links	Same as expected result	Pass
TC-T07	Dashboard Navigation	1. Click the "Members" tab. 2. Click the "Dashboard" tab.	Correct pages open without breaking the session	Same as expected result	Pass
TC-T08	View Profile	Click the "Profile" tab	System returns trainer profile including name, email, phone, specialty, join_month	Same as expected result	Pass
TC-T09	Session Persistence	1. Login as Trainer. 2. Refresh the browser page.	User remains logged-in	Same as expected result	Pass
TC-T10	Create Session	1. Navigate to Sessions. 2. Input valid session data (Title: Strength Basics Date: 2025-01-30 Time: 10:00 AM). 3. Click "Create".	Session created successfully	Same as expected result	Pass
TC-T11	View Reports	1. Navigate to Reports. 2. Select a valid date range.	System displays trainer performance KPIs and charts	Same as expected result	Pass
TC-T12	View Reports	1. Navigate to Reports. 2. Select an empty date range.	Empty analytics screen with informational message	Same as expected result	Pass
TC-T13	View Member Management Page	1. Login as a Trainer 2. Click the Members tab	The "Member Management" page loads successfully. Key metrics (Total Members, Active, Avg. Attendance, New This	Same as expected result	pass

			Month) are displayed and correct. A list of all members is displayed, showing (Name, Email, Phone, Status, Join Date, Attendance).		
TC-T14	add member (valid data)	1. Login as a Trainer. 2. Click the Members tab, then the "+ Add Member" button. 3. Fill in the form: - Full Name: "maria" - Email: marial@example.com - Phone: +966 66 000 000 4. Click Add Member.	• The new member appears immediately in the member list with Status: Active, Attendance: 0%, and the correct Name/Email/Phone. • Total Members count increases (e.g., from 5 to 6) and all information is updated correctly	Same as expected result	pass
TC-T15	Add Member (Duplicate Email)	1. Click + Add Member. 2. Fill in the form using an existing email (e.g., marial@example.com). 3. Click Add Member.	Error message displayed: {"error": "Email already exists"}.	Same as expected result	Pass
TC-T16	Search Member (Valid)	1. Login as Trainer. 2. Navigate to Members. 3. Enter part of a member's name or email (e.g., Alreem) into the search bar.	Search results filtered correctly, displaying only the matching member (Alreem).	Same as expected result	Pass
TC-T17	Search Member (No Results)	1. Login as Trainer. 2. Navigate to Members. 3. Enter a non-existent name (e.g., Lamar) in the search bar.	A blank screen is shown (means that no member is found)	Same as expected result	Pass
TC-T18	Logout	Click logout	Session ended; redirected to login and access to the dashboard required login again	Same as expected result	Pass

6.4 Members Test Cases

Table 5: Members Test Cases

Test Case ID	Feature	Input	Expected Output	Actual Result	Pass/Fail
TC-MEM-01	Landing page Load	User opens the FitHub web app	Landing page loads showing logo, app title, and "Get Started / Member Login" buttons	Same as expected result	Pass
TC-MEM-02	Login (valid credentials)	1. Open the login page. 2. Enter Email: testmember@test.com Password: password123 3. Click "Login".	System authenticates members and redirects to member Dashboard, and the Dashboard loads successfully	Same as expected result	Pass
TC-MEM-03	Login (invalid password)	1. Open the login page. 2. Enter Email: testmember@test.com Password: wrong123. 3. Click "Login".	Error 401 Unauthorized	Same as expected result	Pass
TC-MEM-04	Login (invalid email)	1. Open the login page. 2. Enter Email: wrong@example.com Password: password123. 3. Click "Login".	Validation error 400 (Invalid email format)	Same as expected result	Pass
TC-MEM-05	Login (empty field)	1. Open the login page. 2. Leave Email and Password empty. 3. Click "Login".	Client-side validation: required fields	Same as expected result	Pass
TC-MEM-06	Member Dashboard	Click on "Dashboard" tab	System displays member KPIs, upcoming sessions, recent sessions, and quick links.	Same as expected result	Pass
TC-MEM-07	View profile	Click the "Profile" tab	System returns member profile including name, email, phone, achievements, and join date.	Same as expected result	Pass
TC-MEM-08	Log workout	1. Open Log Workouts 2. fill Fields 3. submit	Workout is created successfully.	Same as expected result	Pass
TC-MEM-09	View Workout History	Click "My Workouts"	System lists workouts + summary totals	Same as expected result	Pass

TC-MEM-10	Cancel Session	1.Navigate to My Sessions 2. Select a booked session 3. Click “Cancel”	Sessions is canceled successfully and removed from schedule	Same as expected result	Pass
TC-MEM-11	Edit Profile	1. Navigate to profile 2. Update profile information 3. Click “Save”	Profile data is updated successfully and saved.	Same as expected result	Pass
TC-MEM-12	View Available Sessions	1. Login as Member 2. Navigate to “Sessions”	System displays list of available sessions	Same as expected result	Pass
TC-MEM-13	Book Session	1.Navigate to Session 2. Select available session 3. Click “Book”	Session is booked successfully.	Same as expected result	pass
TC-MEM-14	Book Full Session	1. Select a fully booked session 2. 2. Click “Book”	System prevents booking and displays an error message	Same as expected result	pass
TC-MEM-15	View Progress & Analytics	Click "Progress"	Display performance trends (weekly/monthly/overall), streaks, and activity breakdowns	Same as expected result	Pass
TC-MEM-16	Logout	Click “Logout”	User is logged out and redirected to login page	Same as expected result	Pass

7. Test Data

This section defines the datasets used to execute and validate the test cases in this Software Test Specification (STS).

The test data covers **valid, invalid, boundary, and relational-integrity scenarios** to ensure full coverage of authentication, profile management, session booking, workouts, and trainer operations.

7.1 Admin Test Data

(Used for testing admin authentication and restricted-access scenarios)

7.1.1 Valid Admin Account

Table 6: Valid Admin Account

ID	Full Name	Email	Password
1	Admin User	admin@test.com	Password123

7.1.2 Invalid Admin Credentials

Table 7: Invalid Admin Account

Case	Input	Expected Result
ADM-E01	Email = "admin@" (with any password)	Login is rejected and an error message is shown for invalid credentials.
ADM-E02	Correct email + wrong password	Login is rejected and an error message is shown for invalid credentials.
ADM-E03	Empty fields	Login is rejected and a validation message is shown for required fields.

7.2 Member Test Data

7.2.1 Valid Member Accounts

Table 8: Valid Member Accounts

ID	Full Name	Email	Password	Status	Join Month	Attendance
1	Fatimah Al-Mousa	fatimah@example.com	password123	Active	Dec 2025	%90
2	Test Member	testmember@test.com	password123	Active	Dec 2025	%90

7.2.2 Invalid Member Accounts

Table 9: Invalid Member Accounts

Case	Input	Expected Result
M-E01	Email = "fatimah@" (with any password)	Login is rejected and an error message is shown for invalid credentials.
M-E02	wrong password	Login is rejected and an error message is shown for invalid credentials.
M-E03	Duplicate email	Registration is rejected and an error message indicates that the email is already in use.

7.3 Trainer Test Data

Table 10: Valid Trainer

ID	Full Name	Email	Password	Specialty	Join Month	Status
1	Test Trainer	testtrainer@test.com	password123	General Training	Dec 2025	Active

2	Zainab Almousa	zainab.almousa@fithub.com	password123	yoga	Dec 2025	Active
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7.3.1 Valid Trainer

7.3.2 Invalid Trainer Inputs (Admin / Trainer creation)

Table 11: Invalid Trainer Input

Case	Input	Expected Result
T-E01	Missing required fields (e.g., name/email)	Trainer creation/update is rejected
T-E02	Email not unique / Duplicate email	Trainer creation is rejected

7.4 Session Test Data

7.4.1 Valid Session

Table 12: Valid Session

Id	Title	Date	Time	Location	Duration	Capacity	Booked
1	HIIT Training	2025-01-25	10:00 AM	Studio A	45 min	12	8
2	Strength & Conditioning	2025-01-26	2:00 PM	Gym Floor	60 min	15	12
3	Boxing Fundamentals	2025-01-28	5:00 PM	Boxing Ring	45 min	8	6

7.4.2 Invalid Session

Table 13: Invalid Session

Case	Input	Expected Result
S-E01	Missing required field (e.g., title/date)	Session is not created and a validation message is shown on the form.

7.5 Booking Test Data

7.5.1 Valid Booking

Table 14: Valid Booking

Case	Member_id	Session_id	Expected
B01	1	1	Booking Created
B02	2	2	Booking Created

7.6 Workout Test Data

7.6.1 Valid Workout

Table 15: Valid Workout

ID	Member_id	Activity	Duration	Calories	Date	Streak
W01	1	HIIT	45	300	2025-01-25	14
W02	2	Strength	60	250	2025-01-26	10

7.6.2 Invalid Workout

Table 16: Invalid Workout

Case	Input	Expected Result
W-E01	Missing requires Activity or Duration.	Workout is not added and validation messages are shown.
W-E02	Empty form submitted.	Workout is not added and required-field messages are shown.

7.7 Achievement Test Data

7.7.1 Valid Achievement

Table 17: Valid Achievement

Ach ID	Member_id	Description	Title	Date	Icon
1	1	Completed 7 consecutive workout days	7 Day Streak	2025-01-25	
2	2	Maintained 90% attendance rate	Consistency Star	2025-01-30	

7.7 Attendance Test Data

(Attendance is reflected in reports and percentages, not as a directly editable entity in the UI.)

7.7.1 Sample Attendance Data Used for Reports

Table 18: Sample Attendance Data Used for Reports

ID	Sessions Attended	Total Sessions	Attendance %	Status	Date
1	9	10	90%	Present	2025-01-25
2	8	10	80%	Present	2025-01-26

These values are used to verify that attendance percentages and related KPIs are displayed correctly in the Trainer Reports and Member Progress views.

8. Test Environment

8.1 Software

- **Operating Systems:**
 - Windows 10/11 or Linux.
- **Development Tools:**
 - Visual Studio Code.
 - System terminal / command line (for running frontend and backend services)
- **Frontend:**
 - React-based web application
 - Node.js and npm for running the development server
- **Backend:**
 - Python 3.x
 - Flask web framework
- **Database:**
 - SQLite.
- **Web Browser:**
 - Google Chrome (used to access and test the system via localhost)

8.2 Hardware

- A standard PC or laptop capable of running the development environment and browser, with:
 - Stable internet connection (for installing dependencies and tools)
 - Sufficient RAM and storage to run the frontend, backend, and database locally.

9. Responsibilities & Schedule

9.1 Responsibilities

The purpose of assigning testing responsibilities is to ensure that all parts of the FitHub system are fully covered and that each team member has clear ownership of specific testing tasks.

- Zainab Almousa & Alreem Alsadiq – Backend and system testing:
 - Responsible for testing APIs, authentication, sessions, workouts, Admin features, validation, and error handling).
- Lamar & Maria & Lamees – Frontend testing:
 - Responsible for testing the user interface, page layouts, and navigation flows, ensuring that the implemented pages match the Figma prototype and behave correctly on the client side.
- Fatimah – Database testing:

- Responsible for verifying the database schema, relationships, and constraints, checking data integrity, and preparing the test data needed for executing the test cases.
- Lamees – Test documentation:
 - In addition to frontend testing, Lamees is responsible for maintaining the test documentation (STS), recording test results, and logging defects, as well as coordinating re-testing after fixes.

Each member is responsible for executing the assigned tests, reporting any issues found, and re-testing the affected parts of the system after fixes are applied.

9.2 Testing Schedule

Table 19: Testing Schedule

Task	Date
Develop and review test cases	06/12/2025
Prepare test data and test environment	07/12/2025
Execute testing activities (unit, integration, and system testing)	09/12/2025
Troubleshoot and analyze errors discovered during testing	10/12/2025
Modify the system according to the detected errors	11/12/2025
Perform regression testing after fixes	12/12/2025
Finalize test documentation (STS)	13/12/2025

10. Deliverables

10.1 Test Cases Document (Complete Test Case Set)

- Contains all system test cases (Admin, Member, Trainer, Authentication).
- Each test case includes preconditions, inputs, detailed steps, expected results, and final Pass/Fail status.

10.2 Test Execution Report

- Provides a summary of all passed and failed tests.

- Includes screenshots of important test results and successful execution confirmations.
- Documents notes regarding any unexpected behavior or environment specifics.

10.3 Defect Log

- A detailed list of all discovered defects is maintained.
- Includes the severity level and step-by-step reproduction steps for each defect.
- Tracks the affected modules (Admin / Member / Trainer) and the current fix status.

10.4 Test Data Sheets

- Contains the sample accounts (Admin, Trainer, Member) used for execution.
- Includes pre-populated data for Workout, Session, and Achievement scenarios.

11. Appendices

11.1 Glossary of Terms

Table 20: Glossary of Terms

Term	Definition
STS	Software Test Specification document for defining tests and data.
SRS	Software Requirements Specification defining system requirements.
SDD	Software Design Description defining architecture and structure.
Test Case	A structured scenario containing inputs, expected and actual outputs.
Test Data	Sample records used to run and verify test cases.
Black-Box Testing	Testing based on inputs/outputs without viewing source code.
Regression Testing	Re-testing system after fixes to ensure no new issues appear.
Validation	Checking correctness of input data before storing in DB.
RBAC	Role-Based Access Control (Trainer, Member, Admin).
KPI	Key Performance Indicator shown on dashboards (e.g., Attendance %, Sessions This Week).
CRUD	Create, Read, Update, Delete – core operations in database.

11.2 References

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